Example 5.24

Find the equation of the hyperbola with vertices $(0,\pm4)$ and foci $(0,\pm6)$.

Solution

From Fig. 5.38, the midpoint of line joining foci is the centre $C\,(0,0)$.

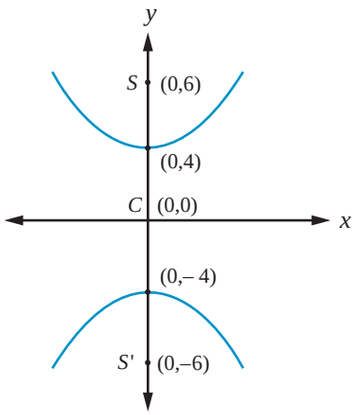


Fig.5.38

Transverse axis is y-axis

$$AA' = 2a \Rightarrow 2a = 8,$$

$$SS' = 2c = 12, c = 6$$

$$a = 4$$

$$b^2 = c^2 - a^2 = 36 - 16 = 20.$$

Hence the equation of the required hyperbola is $\frac{y^2}{16} - \frac{x^2}{20} = 1.$

Example 5.25

Find the vertices, foci for the hyperbola $9x^2 - 16y^2 = 144$.

Solution

Reducing $9x^2 - 16y^2 = 144$ to the standard form,

we have, $\frac{x^2}{16} - \frac{y^2}{9} = 1.$

With the transverse axis is along x-axis vertices are $(-4,0)$ and $(4,0)$;

and $c^2 = a^2 + b^2 = 16 + 9 = 25$, $c = 5$.

Hence the foci are $(-5,0)$ and $(5,0)$.

Example 5.26

Find the centre, foci, and eccentricity of the hyperbola $11x^2 - 25y^2 - 44x + 50y - 256 = 0$

Solution

Rearranging terms in the equation of hyperbola to bring it to standard form,

we have, $11(x^2 - 4x) - 25(y^2 - 2y) - 256 = 0$

$$11(x-2)^2 - 25(y-1)^2 = 256 - 44 + 25$$

$$11(x-2)^2 - 25(y-1)^2 = 275$$

$$\frac{(x-2)^2}{25} - \frac{(y-1)^2}{11} = 1.$$

Centre $(2,1)$, $a^2 = 25, b^2 = 11$

$$c^2 = a^2 + b^2$$

$$= 25 + 11 = 36$$

Therefore, $c = \pm6$

and $e = \frac{c}{a} = \frac{6}{5}$ and the coordinates of foci are $(8,1)$ and $(-4,1)$ from Fig. 5.39.

12th_Maths_EM_Vol1_CH 5_Analytical Geometry.indd 195 16-12-2022 13:32:15